

Apurva Mandalika

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SUMMARY

AI Engineer / Data Scientist with 5+ years of experience building machine learning systems, RAG pipelines, LLM-powered applications, AI agents, copilot-style assistants, and backend AI services using Python, FastAPI, Docker, and cloud tools. Recently completed a Master's in Computer Science from Texas A&M University, with project and professional experience across GenAI, predictive modeling, customer analytics, document Q&A workflows, and enterprise-facing AI solutions.

SKILLS

Technical: Python, SQL, pandas, NumPy, scikit-learn, PyTorch, Sentence Transformers, predictive modeling, feature engineering, data mining, data cleaning, data wrangling, EDA, A/B testing, statistical analysis, customer analytics, model evaluation, model performance monitoring, forecasting, RAG, prompt engineering, hybrid retrieval, semantic search, reranking, LLM routing, REST APIs, **Tools:** Gemini API, Claude, Tableau, Excel, Hive, Hadoop, FastAPI, Flask, Streamlit, Docker, AWS, GCP, Git, GitHub.

EXPERIENCE

Senior Consultant – Data Scientist, Heartland Community Network

Sep 2025 – Aug 2026

- Built a Claude API-powered document Q&A assistant deployed on Streamlit Cloud, enabling clients to upload documents and get instant AI-generated summaries and answers in plain English.
- Evaluated response quality across 20+ document QA pairs achieving ~91% answer accuracy, demonstrating reliability for business document workflows.
- Followed Agile practices across client engagements - gathering requirements, delivering in iterative phases, incorporating client feedback, and holding internal progress reviews between sprints.

Senior Data Scientist, [24]7.ai

Jun 2022 – Jul 2023

- Developed AR-based solutions for iOS and Web, improving resolution rates by 50% compared to traditional methods.
- Led 15+ POCs for an AR feature, securing a patent with USPTO - PCT/IB2023/050635.
- Collaborated with cross-functional teams to integrate AR solutions into the company platform.

Data Scientist, [24]7.ai

Jul 2020 – May 2022

- Introduced and implemented AR solutions using ARKit and computer vision to develop video-based customer support.
- Performed customer analytics reporting, generating ad hoc reports in Excel for stakeholders, delivering 10+ reports per month.
- Built and maintained model performance dashboard in Excel using Hive, SQL, and Python to monitor KPIs, performance trends, and business impact for 23 client predictive targeting models.
- Increased actionable insights by 40% by consolidating reporting and dashboard efforts.
- Created comprehensive JIRA-Confluence documentation for models, improving transparency and accelerating onboardings.

Analytics Consultant, [24]7.ai

Jan 2019 – Jun 2020

- Created Time on Page (TOP) and Page-Level Propensity to Purchase after Chat (P2PC) models, increasing chat propensity by 8%, RPC by 6%, and conversion rates by 12%.
- Deployed scalable predictive targeting models for 4 clients using Hadoop and FlashML for customer behavior modeling.
- Conducted detailed EDA, preprocessing, anomaly filtering, and A/B testing, improving model performance by nearly 5%.

PROJECTS

Agentic RAG System with MCP Tool Orchestration | Tools: Python, Anthropic MCP, Claude API, FAISS, BM25, FastAPI.

- Architected an agentic RAG pipeline using Anthropic's MCP where a Claude-powered agent autonomously routes queries across hybrid document retrieval (FAISS + BM25), live web search, and structured data lookup via registered MCP tool servers with cross-encoder reranking.
- Evaluated across a 25-document corpus with 15 diverse test queries, achieving 93%+ retrieval coverage with multi-source reasoning and end-to-end source attribution in every response.

RetailIQ AI: Retail Knowledge Assistant | Tools: Python, RAG, FAISS, BM25, cross-encoder, Gemini API, FastAPI.

- Built a production-style retail knowledge assistant using RAG to answer questions from pricing, promotions, returns, and store operations documents, implementing a hybrid retrieval pipeline (FAISS semantic search + BM25 + cross-encoder reranking) for policy-heavy queries with grounded source citations.
- Evaluated pipeline quality using both supervised (100% source retrieval, ~89% answer coverage across 15+ QA pairs) and unsupervised methods (embedding-based cosine similarity via Sentence Transformers to score retrieval relevance and answer groundedness without labeled ground truth).

Walmart Demand Forecasting Copilot | Tools: Python, Forecasting, LLM Routing, FastAPI, XGBoost, Streamlit.

- Developed an AI-powered retail analytics assistant that uses Gemini-based routing to interpret natural language queries and dynamically invoke forecasting or KPI tools enabling a lightweight agent-style workflow for demand planning.
- Built and served a weekly demand forecasting pipeline with XGBoost, lag features, and rolling-window features, integrating predictions and KPI summaries into a FastAPI + Streamlit application for interactive analytics.

EDUCATION

Texas A&M University, Master's in Computer Science (CGPA: 3.9/4)

Aug 2023 – May 2025

Amrita Vishwa Vidyapeetham, Bachelor's in Computer Science and Engineering (CGPA: 9.35/10)

Aug 2015 – May 2019